

Exp. No. 4.3

Design and Simulation of 4-bit Binary Adder using Verilog code in Xilinx

Aim

To design and simulate a 4-bit binary adder using Verilog HDL and verify its functionality using the Xilinx Vivado simulator.

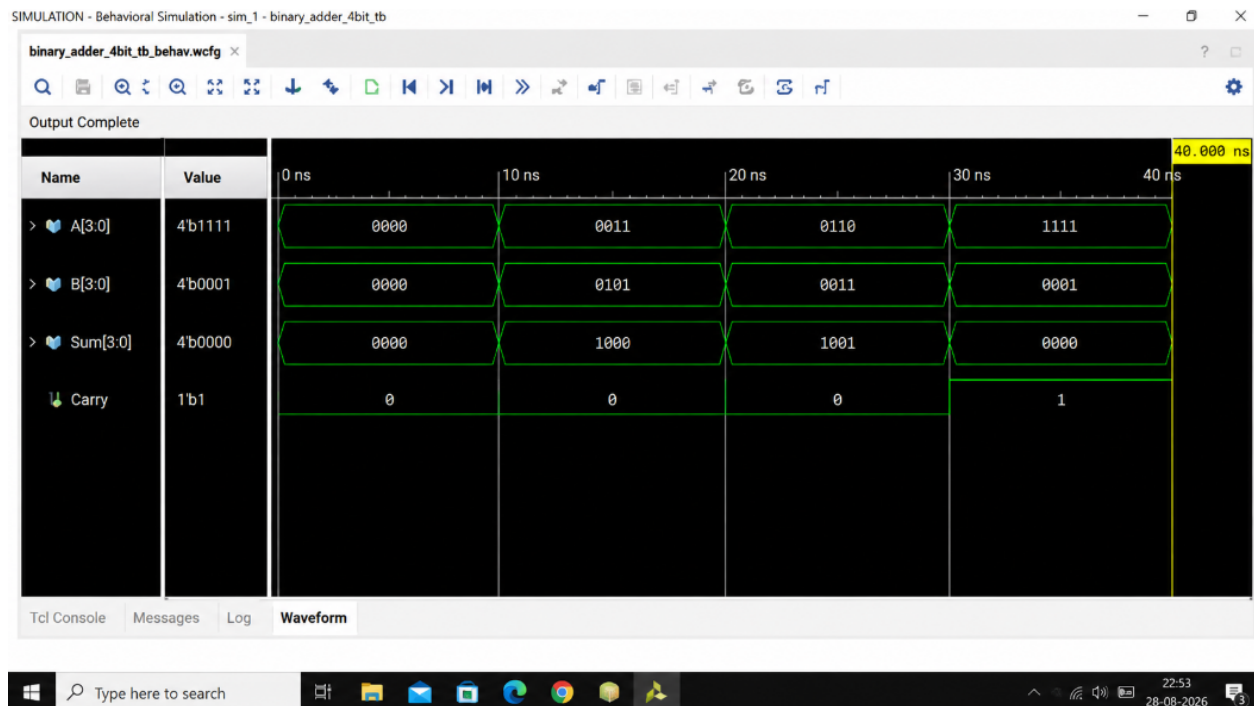
Verilog Code

```
module binary_adder_4bit(  
    input [3:0] A,  
    input [3:0] B,  
    output [3:0] Sum,  
    output Carry  
);  
    assign {Carry, Sum} = A + B;  
endmodule
```

Testbench

```
`timescale 1ns / 1ps  
module binary_adder_4bit_tb;  
    reg [3:0] A;  
    reg [3:0] B;  
    wire [3:0] Sum;  
    wire Carry;  
  
    binary_adder_4bit uut (  
        .A(A),  
        .B(B),  
        .Sum(Sum),  
        .Carry(Carry)  
    );  
  
    initial begin  
        A = 4'b0000; B = 4'b0000; #10;  
        A = 4'b0011; B = 4'b0101; #10;  
        A = 4'b0110; B = 4'b0011; #10;  
        A = 4'b1111; B = 4'b0001; #10;  
  
        $finish;  
    end  
endmodule
```

Output



Result: The 4-bit binary adder was designed and simulated successfully in Xilinx Vivado. For A=0000,B=0000 the sum was 0000 with carry 0; for A=0011,B=0101 the sum was 1000 with carry 0; for A=0110,B=0011 the sum was 1001 with carry 0; for A=1111,B=0001 the sum was 0000 with carry 1, matching the expected results.